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## **Micro-Proppant Transport in 3D-Printed Rough Natural Fractures Using Time-Resolved Particle Image Velocimetry**

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### **Abstract**

Unconventional oil and gas reservoirs contain numerous secondary natural fractures ( $<1$  mm), making it difficult for conventional-sized proppants to enter these fractures, resulting in a low effective stimulated reservoir volume. Micro-proppants (particle size  $<200$  mesh) have the potential to enter micron-scale fractures, activate secondary fracture networks, and enhance hydrocarbon recovery. However, limited experimental studies have visualized the proppant transport behavior in fractures with dual roughness on both surfaces.

In this study, a Time-Resolved Particle Image Velocimetry (PIV) based experimental setup was employed to investigate the transport and placement of fluorescent-coated micro-proppants in rough fractures. Synthetic fracture surfaces with controlled fractal dimensions and RMS roughness values were generated using SynFrac software and fabricated via 3D printing. A systematic investigation was conducted to assess the effects of proppant concentration, injection rate, and fracture dip angle on micro-proppant placement. Pressure sensors were used to quantify micro-proppant transport dynamics in rough secondary fractures and to identify key parameters affecting placement efficiency.

Experimental results show that increasing fracture inclination significantly influences micro-proppant transport and deposition. As the fracture angle increases, the deposition pattern shifts from cluster-based blockage to dune-like accumulation, reducing placement efficiency in the distal region. Higher proppant concentrations enhance interparticle interactions, leading to aggregation and bridging, which limit transport distance and hinder effective placement in narrow fracture sections. Increasing injection rate promotes deeper transport of micro-proppant. However, when the rate exceeds a critical threshold ( $Re = 16.42$ ), excessive proppant migration toward the fracture tip reduces deposition near the inlet, resulting in uneven distribution. Therefore, the interaction between proppant concentration and injection rate jointly controls placement patterns and efficiency. Optimizing these parameters can significantly improve micro-proppant distribution and placement performance within the fracture.

These findings provide new insights into micro-proppant transport and placement in rough-walled fractures, contributing to the optimization of stimulation strategies for unconventional reservoirs.

**Keywords:** micro-proppant, PIV, proppant transport, reservoir fracturing, rough fracture

## Introduction

The decline of conventional hydrocarbon reserves has redirected energy development toward unconventional reservoirs, a shift primarily driven by advances in hydraulic fracturing technology (Liao et al., 2024; Tosta et al., 2023). This technique is widely used to create fracture networks in low-permeability formations (Qi et al., 2020). Proppants play a key role in maintaining fracture conductivity (Liu et al., 2020). However, field and lab studies show that fracturing often generates secondary natural fractures with apertures between 30 and 1000  $\mu\text{m}$  (Gale et al., 2014). These narrow fractures are common in unconventional plays but are poorly propped due to the size of traditional proppants ( $\geq 100 \mu\text{m}$  or 150 mesh), which cannot enter or stay in such tight spaces (Walton and McLennan, 2013). To improve stimulation efficiency, micro-sized proppants (MPs) ranging from 100 nm to 100  $\mu\text{m}$  have been proposed. MPs—such as silica flour, ceramics, fly ash, and other industrial byproducts—can access narrower fractures and offer a cost-effective alternative (Li et al., 2024). Among them, fly ash has attracted interest due to its low cost, favorable particle size distribution, and comparable conductivity to ceramics. Surface modification agents (ASMAs) are often used to improve MP transport by reducing aggregation, settling, and wall adhesion, enhancing proppant placement in complex fracture systems (Aslannezhad et al., 2021). Lab tests show that MPs, especially when combined with ASMAs, can significantly increase fracture conductivity (Tian et al., 2023; Kim et al., 2019). MPs also tend to distribute more uniformly in rough or tortuous fractures, further improving flow capacity (Li et al., 2024; Knez et al., 2020; Zhou et al., 2023). Field applications in formations like the SCOOP Woodford and Bakken confirm these results, showing enhanced when MPs are used in fracturing treatments (Li et al., 2024). 1

Experimental and numerical studies have substantially advanced the understanding of two-phase particle-laden flows that increased roughness accelerates settling and shortens horizontal transport distance, promoting early blockage. CFD-DEM simulations further show that higher flow rates extend transport length but reduce equilibrium bed height (Zhou et al., 2024). As roughness increases, equilibrium height rises slightly but remains below that of smooth fractures, while transport length decreases but still exceeds that in smooth cases. Tatman et al. (2022) observed that larger proppants settle faster in rough fractures due to wall effects. Despite recent progress, most studies still focus on vertical fractures, and proppant transport behavior in rough horizontal fractures remains poorly understood.

Particle Image Velocimetry (PIV) is an effective technique for capturing particle motion and has made significant progress in flow testing. It measures fluid velocity through particle images (Hain and Kähler, 2007; Lavoie et al., 2007). In recent years, many studies have applied this method to investigate proppant migration patterns and draw important conclusions. Zhang et al. (2024) used PIV for two-phase velocity testing, employing exposure differences to separate tracers and track single-phase velocity. They summarized two dune growth patterns: "overall longitudinal growth" and "growth along the fracture length," and quantitatively studied the underlying micro-motion mechanisms. Guo et al. (2022) applied PIV/PTV algorithms to calculate two-phase velocity distributions, finding that increased flow rate, fracture width, and viscosity led to a larger area influenced by vortices, resulting in proppant removal. (Liu et al., 2020) showed that the formation process is similar across different fracturing fluids and can be divided into four stages: initial dune formation, growth, equilibrium state establishment, and piston-like movement. The dune's equilibrium height, build-up angle, advancing angle, and void space are influenced by fluid erosion, vortices, and sand-carrying ability, with its shape primarily depending on fluid velocity and proppant. (Fjaestad and Tomac, 2019) highlighted and quantified the role of particle size in a narrow slot on various flow regimes.

To address the lack of visualized understanding of micro-proppant transport in rough-fractures, this study employs a time-resolved PIV system to experimentally investigate proppant dynamics in fractures with dual-surface roughness. Synthetic fracture models with controlled fractal geometries were fabricated using 3D printing. The effects of injection rate, proppant concentration, fracture roughness, and dip angle on transport and placement behaviors were systematically analyzed. Image segmentation techniques were used to quantify transport efficiency and identify key controlling parameters. This work provides quantitative insights into micro-proppant behavior in realistic rough fractures and offers guidance for optimizing stimulation in unconventional reservoirs.

## Experimental system

### Rough fracture models

The surface morphology of rough secondary fractures is a critical factor controlling proppant transport and placement. In actual fracturing operations, these secondary fractures are not only rough but also randomly oriented. Fig. 1 illustrates the model generation workflow. Synthetic surface data with defined fractal dimensions and root mean square (RMS) roughness (surface height and fracture aperture) were generated using the Brownian method in SynFrac (Glover et al., 1998). A Python script was developed to convert the simulated rough surface into a geometry file (STEP format). The geometry was then imported into modeling software for design and fabricated using 3D printing with transparent photosensitive resin. The rough fracture section in the model has a length of 50 mm, a width of 25 mm, and an average fracture height of 0.25 mm.

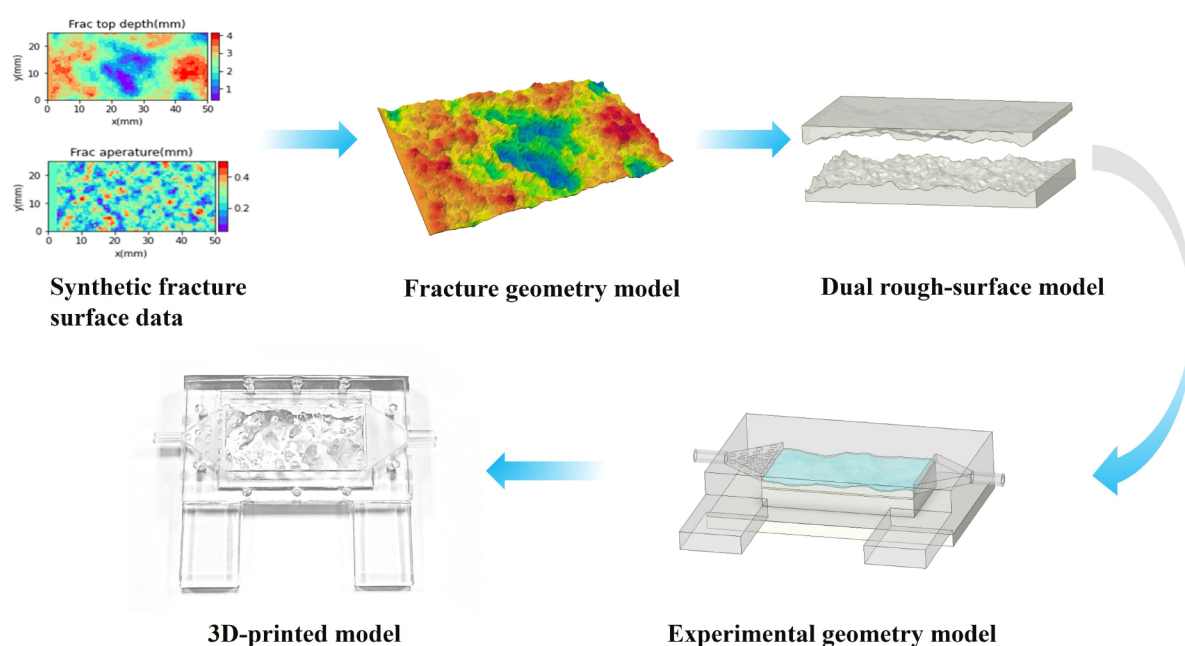


Figure 1—Fracture model generation workflow

### Experimental set-up

Based on the generated rough fracture model, a low-pressure experimental setup was developed to simulate the transport behavior of micro-proppants within rough fractures. A schematic of the overall setup is shown in Fig. 2 a, and the actual experimental device is presented in Fig. 2 b. The system consists of a rectangular rough fracture model, a Harvard pump, a magnetic stirrer tank, pressure sensors, a digital camera, a PIV system connected to a computer for image analysis, a laser light guide arm, a rotatable fixture,

and tubing. The camera is positioned vertically above the model to record videos and capture images during experiments, aiming to track micro-proppant movement. The laser light guide arm is fixed at a  $45^\circ$  angle to the model surface. A planar lens is used as a surface light source to ensure uniform illumination across the rough fracture surface. The rough fracture model is mounted on a rotatable fixture, allowing adjustment of the fracture inclination angle. This setup enables the measurement of velocity fields and the observation of micro-proppant deposition within the entire rough fracture.

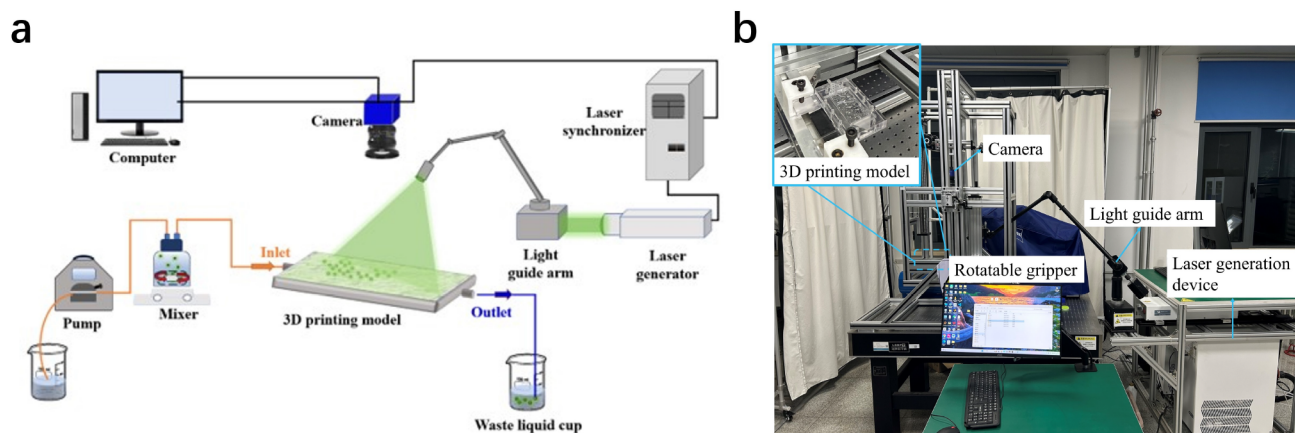


Figure 2—Schematic diagram of the experimental system

## Experimental Procedures

The particles used in the experiment were solid, spherical beads coated with Rhodamine B, with a diameter range of 20–50  $\mu\text{m}$ . To eliminate ambient light interference, a 540 nm cutoff filter was placed in front of the camera lens. Silicone tubing with an outer diameter of 5 mm and an inner diameter of 3 mm was used for the flow lines. Given that micro-proppants are designed to support natural microfractures and secondary fractures in field applications—where flow velocities are typically below 0.1 m/s (Alotaibi and Miskimins, 2019)—the inlet velocity of the fluid was set between 7.5 and 12.5 ml/min during the experiment. Based on field practices (Aslannezhad et al., 2021; Danso et al., 2021), the micro-proppant volume concentration was maintained from 0.3 % to 1.5%. Detailed experimental parameters are listed in table 1.

Table 1—Fluid Properties and Experimental Conditions

| Parameter                        | Value              | Descriptions                                   |
|----------------------------------|--------------------|------------------------------------------------|
| $L$ (mm)                         | 50                 | Fracture model length                          |
| $W$ (mm)                         | 25                 | Fracture model width                           |
| $h$ (mm)                         | 0.25               | Fracture model average height                  |
| $Q$ (ml/min)                     | 7.5, 10, 12.5      | Injection rate                                 |
| $Re$                             | 9.85, 13.14, 16.42 | Dimensionless injection rate (Reynolds number) |
| $V$ (m/s)                        | 0.02, 0.027, 0.033 | flow rate                                      |
| $C$ (%)                          | 0.375, 0.75, 1.5   | Micro-proppant volume concentration            |
| $\theta$ ( $^\circ$ )            | 0, 45, 90          | Fracture orientation angle                     |
| $\mu_p$ ( $\mu$ )                | 20–50              | Micro-proppant size                            |
| $\rho_p$ ( $\text{g/cm}^3$ )     | 2.1                | Micro-proppant density                         |
| Temperature ( $^\circ\text{C}$ ) | 25                 | -                                              |



To conduct micro-proppant visualization experiments, the model and flow lines were first cleaned using a high-pressure air dusting tank. This step also minimized the formation of uneven wet areas caused by residual water droplets on the inner wall of silicone tubing, which could act as nucleation sites for bubble attachment during water injection. After the model and flow lines were connected in sequence, a Harvard pump was activated and set at the same injection rate as the experiment. Distilled water was continuously injected to displace air bubbles from the system. Once fully saturated with distilled water, the model and tubing were mounted on a rotating platform. The camera was then turned on. The exposure mode was set to single frame with an exposure time of 10,000  $\mu\text{s}$ . The focal length was adjusted to ensure image clarity, and the model dimensions were calibrated. The camera lens was then covered, and the laser intensity and position were adjusted. Exposure time was reset to 100  $\mu\text{s}$  before uncovering the lens to complete the PIV setup. Next, the magnetic stirrer in the mixing tank was activated to ensure uniform micro-proppant distribution. Once mixed, the valve was opened, and the PIV system started recording. Simultaneously, the pressure sensors were activated. A total of seven experimental runs were conducted following this procedure.

## Results

In order to quantify the flow and transport dynamics of micro-proppant transport in rough secondary fractures, several parameters are proposed as follows: **The Reynolds number** (Re) is calculated using the following formula:

$$\text{Re} = \frac{\rho_f \bar{u}_f \bar{w}}{\mu_f} \quad (1)$$

Where  $\rho$  is the fluid density ( $\text{kg}/\text{m}^3$ ),  $v$  is the fluid velocity ( $\text{m}/\text{s}$ ),  $\bar{w}$  is the characteristic length (m), and  $\mu$  is the dynamic viscosity of the fluid ( $\text{Pa}\cdot\text{s}$ ).

**The dimensionless injection time** is the time required to displace a fracture volume (FV):

$$FV = \frac{V_f}{Q} = \frac{L\bar{w}h}{u\bar{w}h} = \frac{L}{u} \quad (2)$$

Where  $V_f$  is the fracture volume ( $\text{m}^3$ ),  $Q$  is the flow rate ( $\text{m}^3/\text{s}$ ),  $\bar{w}$  is the fracture width (m),  $\mu$  is the dynamic viscosity of the fluid ( $\text{Pa}\cdot\text{s}$ ),  $h$  is the fracture height (m), and  $L$  is the fracture length (m).

### Effect of injection flow rate

The particle movement is mainly driven by fluid drag force, thus making flow velocity (Reynolds number) as a key factor to control the micro-proppant transports. Fig. 3 shows the transport and settling behavior of micro-proppants in rough secondary fractures at three typical injection flow rates ( $\text{Re} = 9.85, 13.14$ , and  $16.42$ ). As the injection flow rate increases, both the transport velocity and spatial distribution of the micro-proppants change significantly. At  $\text{Re} = 16.42$ , micro-proppants disperse rapidly, forming multiple preferential channels throughout the fracture. In contrast, at  $\text{Re} = 9.85$ , micro-proppant settles and accumulates near the fracture inlet, failing to reach the distal end.

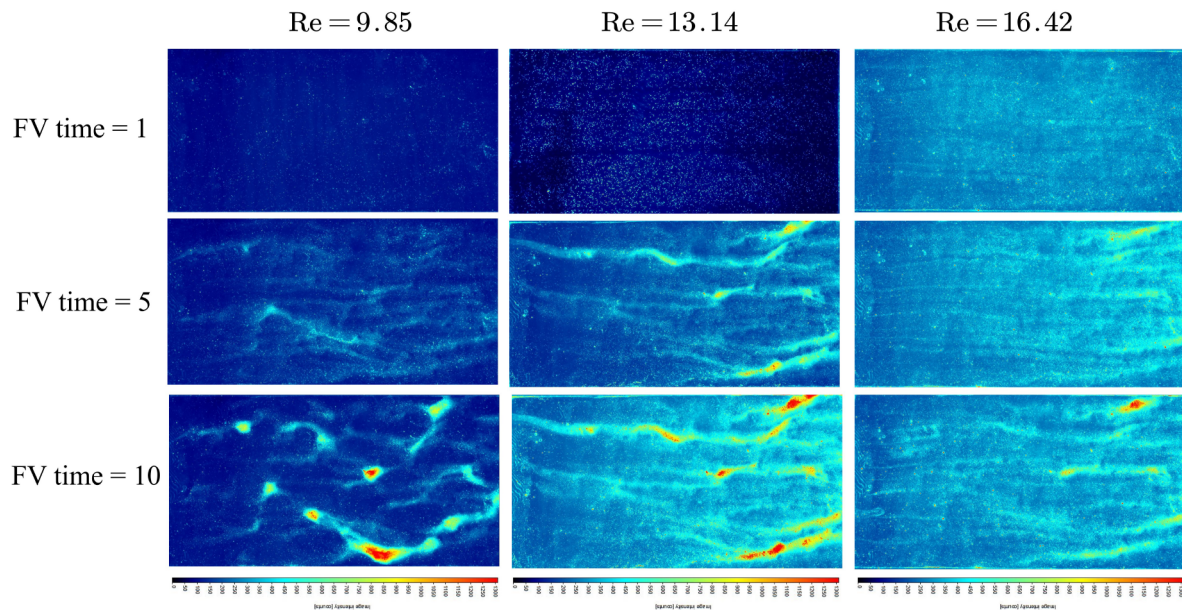


Figure 3—Process of micro-proppant transport and settlement at different injection flow rates

Fig. 4 illustrates the evolution of streamline patterns within rough secondary fractures under varying injection rates, corresponding to different Reynolds numbers. At the early injection stage (FV time = 1), streamlines appear generally straight, indicating that the flow is approximately laminar and follows the fracture geometry. High-velocity regions are mainly located near abrupt changes in fracture geometry, suggesting that surface roughness significantly influences flow distribution. As injection continues and the Reynolds number increases (Re = 9.85 to 16.42), streamlines become more complex, with evident flow diversion around rough features. At longer injection times (FV time = 5 and 10) and under lower Reynolds numbers (Re = 9.85 and 13.14), multiple micro-proppant settling regions are observed, identified by regions of reduced velocity and particle accumulation. These regions often occur where streamlines diverge or converge, creating localized flow barriers. As a result, high-velocity flow is gradually redirected around these regions, indicating the development of preferential flow paths.

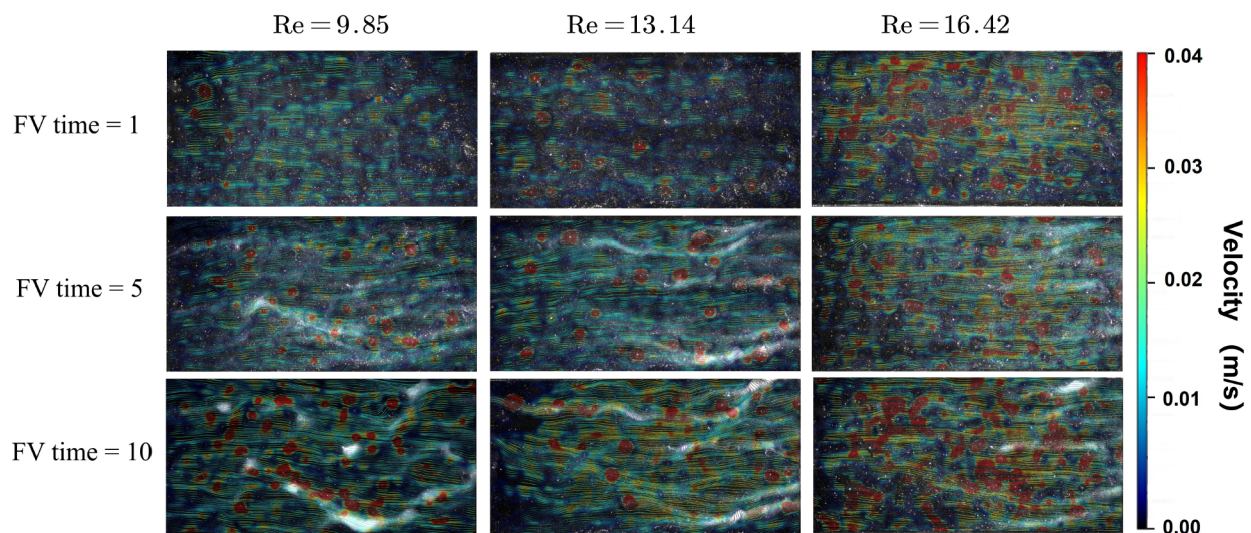


Figure 4—Streamline variations in rough secondary fractures under different injection flow rates

Under higher Reynolds number conditions ( $Re = 16.42$ ), although fracture geometry still affects flow distribution, stronger fluid momentum suppresses particle settling. No distinct deposition regions are formed, and high-velocity regions remain dominant. This suggests enhanced fracture conductivity and minimal disruption of preferential channels by micro-proppants.

To further quantify the transport and settling behavior of micro-proppants within rough secondary fractures, real-time pressure monitoring was conducted during fluid injection. Fig. 5 shows the pressure evolution over time under various injection flow rates, corresponding to different Reynolds numbers. The results indicate that increasing the injection rate significantly accelerates the pressure buildup. The high-flow-rate group consistently exhibits higher pressure values throughout the injection, reflecting a stronger dynamic driving force. Under low flow rate conditions, a single notable pressure surge is observed at approximately 90 s, with a peak of 126.4 mbar. This fluctuation is likely caused by rapid micro-proppant settling near the fracture inlet, leading to temporary blockage and a sudden pressure rise. In contrast, high flow rates result in more frequent and larger pressure fluctuations. At  $Re = 16.42$ , the peak pressure reaches 227.2 mbar, which is 79.74% higher than that of the low-flow-rate case. This suggests that higher flow velocities enhance erosion at fracture junctions, inducing transient flow disturbances and local structural rearrangement, thereby causing repeated pressure oscillations.

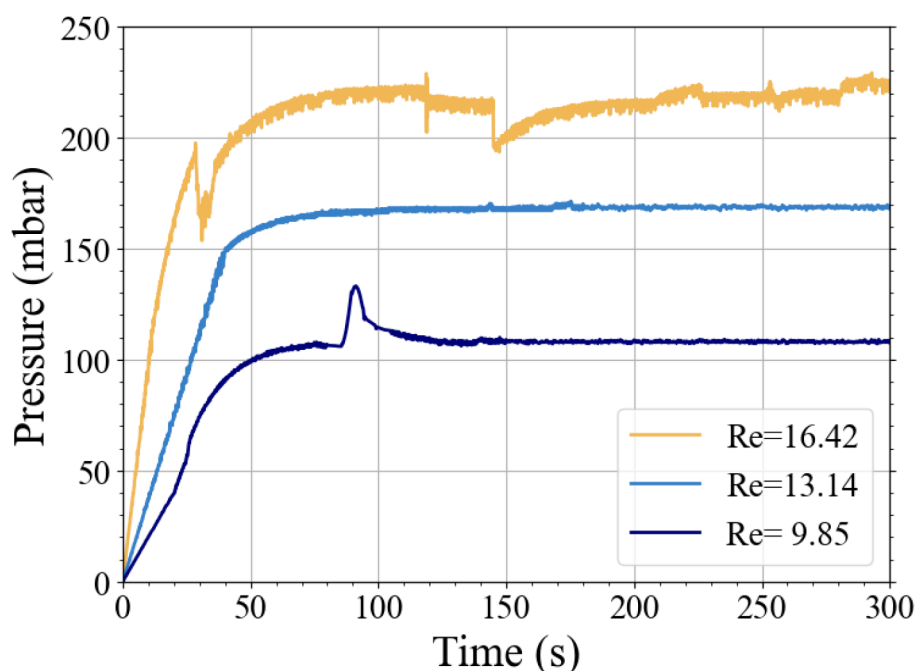


Figure 5—The variation of pressure with the injection flow rates

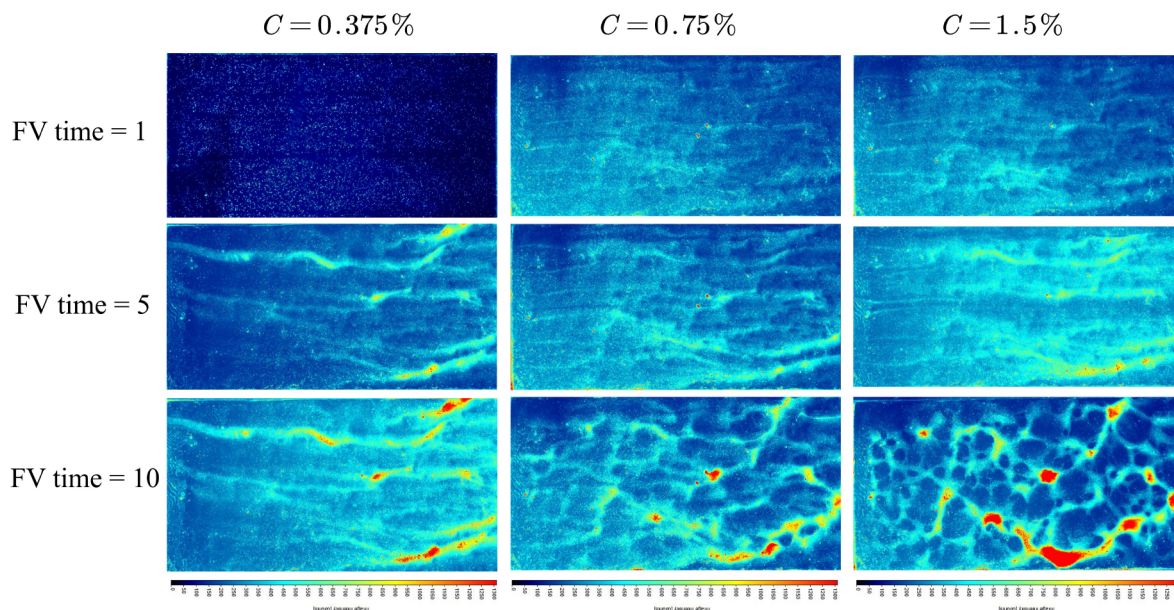
The reduced flow velocity weakens the drag of the fluid in the particles and enhances particle–wall interactions, promoting aggregation. Gravity dominates particle settling and transport. Higher inertial effects delay settling, allowing deeper micro-proppant transport. In contrast, lower injection velocity intensifies aggregation and shortens transport distance, while higher Reynolds numbers lead to deeper and more uniform settlement.

### Effect of injection particle concentrations

As shown in Fig. 6, the deposition rate of micro-proppants in rough secondary fractures increases with higher injection concentrations, reflecting greater particle availability. However, the main settling regions remain nearly unchanged. This is due to the fixed geometry of the fracture, where valley regions act as stable



low- velocity regions that trap micro-proppants. Therefore, higher concentrations lead to more deposition but do not shift the spatial pattern of particle placement.



**Figure 6—Process of micro-proppant transport and settlement at different injection particle concentrations**

Fig. 7 illustrates the evolution of streamline patterns within rough secondary fractures under varying particle concentrations and injection times. At the initial injection stage (FV time = 1), streamlines are relatively straight throughout the fracture, indicating that the flow is not yet significantly disturbed. No apparent micro-proppant deposition or blockage is observed in the rough secondary fracture regions. As particle concentration increases (from 0.375% to 1.5%) and injection continues (FV time from 1 to 10), localized deposition zones gradually form within the fracture, particularly around rough asperities. Streamlines become increasingly distorted and exhibit curved trajectories, suggesting enhanced flow interference due to micro-proppant accumulation. However, high-velocity flow regions remain largely centered along the main flow path, showing limited spatial shift across different concentrations. Notably, at the highest concentration ( $C = 1.5\%$ ) and longest injection time (FV time = 10), severe micro-proppant blockage occurs near the fracture center and outlet. High-velocity flow is confined to the inlet region, significantly reducing effective fluid penetration. This indicates that injecting high-concentration micro-proppant into rough fractures may induce pronounced outlet blockage.



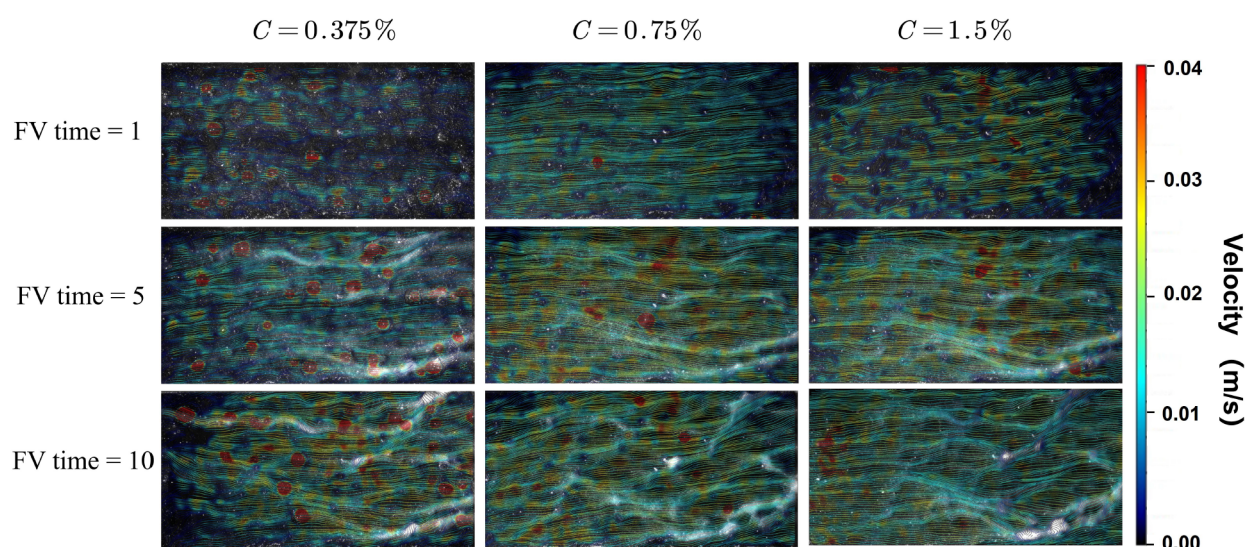


Figure 7—Streamline variations in rough secondary fractures under different injection particle concentrations

Fig. 8 shows the pressure evolution in rough secondary fractures under different proppant concentrations. As the dimensionless injection time (FV) increases, the dimensionless pressure rises for all three concentrations. At 10 FV, the pressures for 0.375%, 0.75%, and 1.5% concentrations are 169.2 mbar, 189.1 mbar, and 254.6 mbar, respectively. Higher proppant loading leads to increased fracture pressure. At 1.5% concentration, pressure rises more rapidly, while 0.375% results in a more moderate increase. This trend is mainly attributed to the formation of larger bridging and blockage zones at higher concentrations, which restrict flow and accelerate pressure buildup.

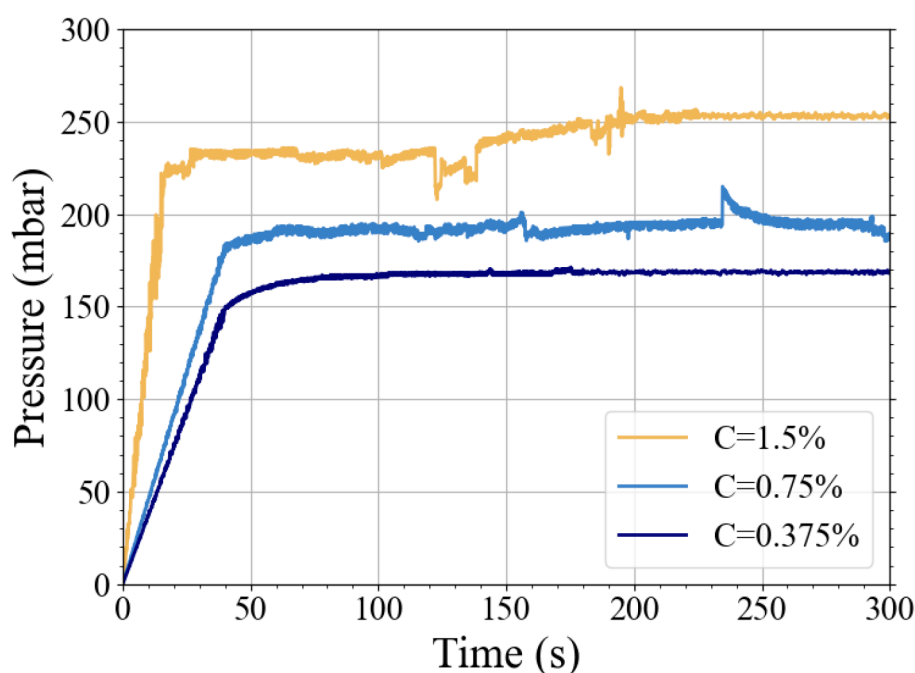


Figure 8—The variation of pressure with the injection particle concentrations

High particle concentration significantly improves micro-proppant settlement efficiency by altering flow behavior through intensified particle–particle and particle–wall interactions. These interactions disturb the streamline structure, promote localized deposition near rough asperities, and lead to severe outlet blockage.

As a result, flow resistance increases and pressure builds up more rapidly, especially in high-concentration cases (e.g., 1.5%), where bridging and clogging zones substantially hinder fluid transport.

### Effect of fracture orientation

In reservoir stimulation, geological factors such as stress distribution, natural fractures, and bedding planes can cause vertical deviation of fracture propagation, resulting in inclined fractures. We conducted experiments at different strike angles:  $0^\circ$  (horizontal fracture),  $45^\circ$ , and  $90^\circ$  (vertical fracture).

Fig. 9 shows the distribution of micro-proppants under three strike angles. The strike angle has a significant effect on micro-proppant transport and deposition in rough fractures. At  $0^\circ$ , micro-proppants tend to form discrete clusters due to in-teraction with rough fracture surfaces. As the strike angle increases, the deposition pattern shifts from cluster-based blockage to dune-like structures, primarily driven by gravity. Notably, no dune-like patterns are observed at  $90^\circ$ , where vertical orientation reduces particle–wall interactions such as collision and friction, allowing proppants to travel more easily toward the fracture tip.

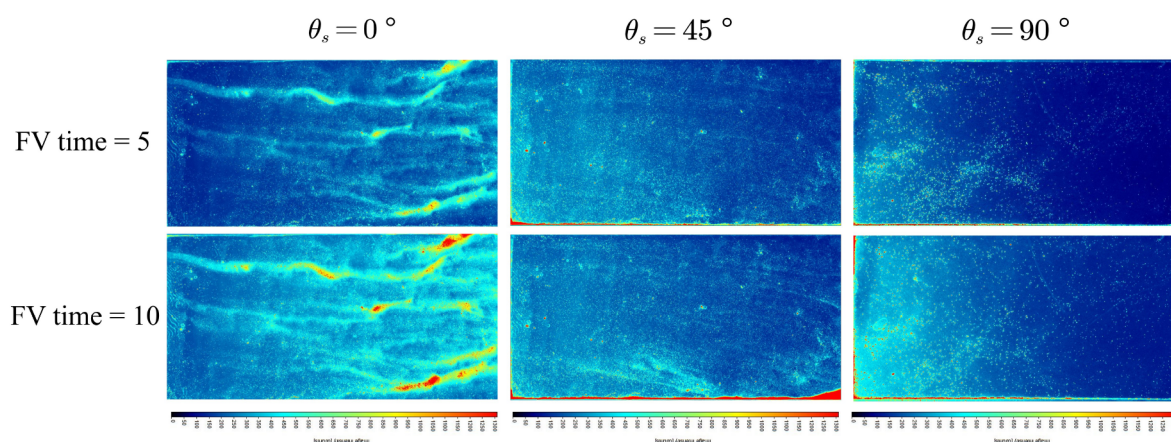


Figure 9—Process of micro-proppant transport and settlement at different injection rates at different fracture orientation

Fig. 10 illustrates the evolution of streamline patterns within rough secondary fractures under two strike angle conditions. Compared to the horizontal case (strike angle =  $0^\circ$ ), the flow paths in the  $45^\circ$  strike angle scenario exhibit a noticeable inclination relative to the horizontal direction. As dune-like deposits form near the bottom of the model, the flowable cross-sectional area becomes narrower, and the proportion of high-velocity regions increases accordingly. Notably, a triangular sandbar develops near the outlet, inducing local flow disturbances and streamline deflection.

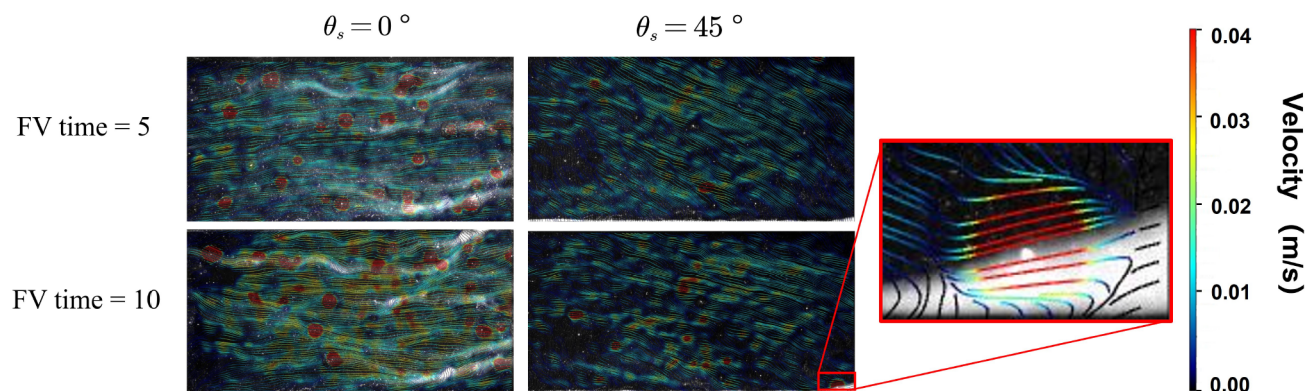


Figure 10—Streamline variations in rough secondary fractures under different fracture orientations

Fig. 11 presents the pressure evolution in rough secondary fracture models under different fracture orientations. As the dimensionless time (FV) increases, the dimensionless pressure rises for all three cases, indicating progressive micro-proppant transport and accumulation. However, the pressure buildup rate decreases with increasing fracture orientation angle. At FV = 10, the measured pressures for fracture orientations of  $0^\circ$ ,  $45^\circ$ , and  $90^\circ$  are 169.2 mbar, 130.4 mbar, and 100.7 mbar, respectively. The higher pressure observed at  $0^\circ$  is attributed to more pronounced micro-proppant bridging and clustering due to increased interaction with the rough fracture surface, which creates localized flow resistance and elevates backpressure. At  $90^\circ$ , the absence of gravitational settling and reduced particle–wall interactions result in minimal proppant accumulation, leading to the lowest pressure response. The  $45^\circ$  case exhibits the highest pressure fluctuation, likely caused by unstable dune formation and intermittent channel blocking, which induce transient flow disruptions within the fracture.

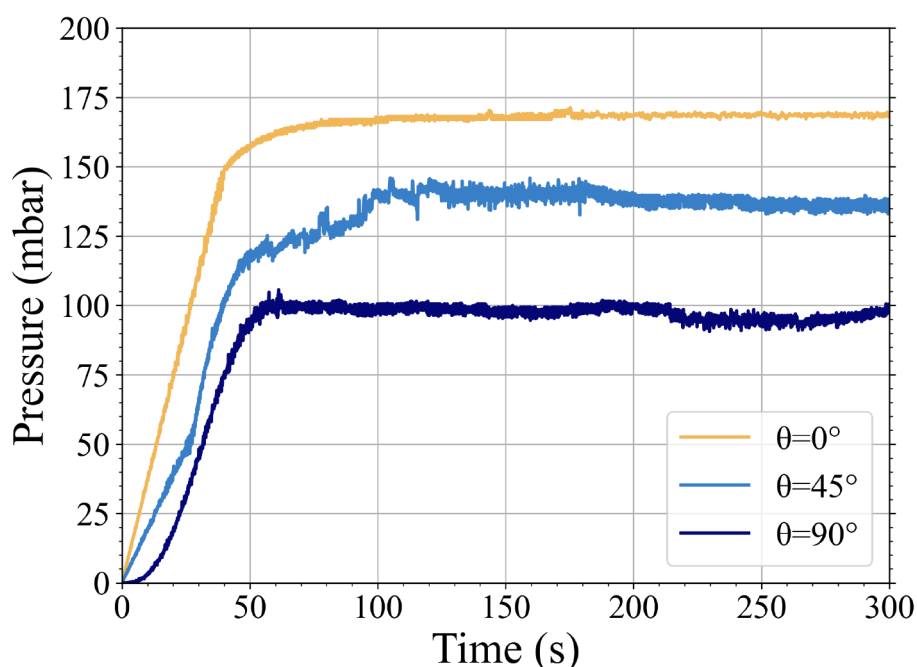


Figure 11—The variation of pressure with the fracture orientation

## Discussion and Conclusions

In this study, we investigated micro-proppant transport in horizontal rough fractures using Particle Image Velocimetry (PIV). The effects of injection flow rate, particle concentration, and fracture orientation were systematically analyzed. We examined the proppant placement patterns, streamline distributions, and pressure variations within the model. In addition, microscopic mechanisms such as channeling and particle–wall interactions were discussed. The key findings are summarized as follows:

- When Re increases from 9.85 to 16.42, the settling zone in the rough secondary fracture model progressively shifts toward the outlet, and the streamlines become straighter and more uniform. Higher flow velocity facilitates the formation of fast, continuous flow paths, improving the transport and placement of micro-proppant and reducing the risk of settling.
- Higher micro-proppant concentrations (up to 1.5%) significantly distort streamlines and create localized deposition, causing outlet blockage and reducing flow paths. Compared to 0.375% concentration, fracture pressure increases by 50% (from 169.2 to 254.6 mbar) after 10 FV of injection. These findings highlight the trade-off between improved settlement efficiency and elevated plugging risk in rough fracture networks.



- Micro-proppant transport and pressure response are strongly influenced by fracture orientation. At  $FV = 10$ , pressure drops from 169.2 mbar ( $0^\circ$ ) to 100.7 mbar ( $90^\circ$ ), with  $45^\circ$  showing peak fluctuations due to unstable dune formation. These results highlight the need to account for fracture geometry in optimizing proppant placement.

These findings support the optimization of micro-proppant design and injection strategies, particularly for applications in rough secondary fractures.

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